Archives of Gerontology and Geriatrics 108 (2023) 104924

Contents lists available at ScienceDirect

Archives of Gerontology and Geriatrics

journal homepage: www.elsevier.com/locate/archger

Review

Effectiveness of virtual reality games in improving physical function, balance and reducing falls in balance-impaired older adults: A systematic review and meta-analysis

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ARTICLE INFO

Keywords:
VR interventions
Old adults
Physical function
Balance
Fall

ABSTRACT

Background: In recent years, sports games based on virtual reality (VR) have been widely used in the prevention and treatment of diseases related to the elderly. However, there seems to be no consensus on the improvement and comparison of physical function, balance and falls in elderly people with balance impairment.

Objective: This study aims to explore the effects of VR intervention on physical function, balance and falls in elderly people with balance impairment.

Methods: Systematic literature searches of the PubMed, Web of Science, Elsevier, Cochrane, CNKI, and Wanfang databases were performed for VR games-related randomized controlled trials or comparison studies among elderly participants with impaired balance, published in English or Chinese until March 20, 2022. The Cochrane collaboration risk of bias tool was used to evaluate the methodological quality of the studies. A meta-analysis was performed to calculate the standardized mean deviation or mean difference of the sample and its 95% confidence interval (CI) in VR games.

Results: The systematic review included 23 studies. The results showed that VR intervention had significant effects on hand grip strength (MD:1.30, $P = 0.040$), knee extension strength (MD:-6.27, $P<0.001$), five times sit-to-stand test scores (MD:1.13, $P = 0.030$), timed up-and-go test scores (MD:-1.01, $P = 0.001$), berg balance scale scores (MD:2.37, $P<0.001$), and falls efficacy scale scores (SMD:-0.28, $P = 0.020$). Subgroup analysis results showed that VR intervention was more effective on improving TUG and BBS scores than the conventional exercise group (MD=-0.54, $P = 0.004$; MD=3.24, $P<0.001$) and the non-intervention group (MD=-0.98, $P = 0.001$; MD=3.30, $P < 0.001$). The balance training-based VR had a significant effect on improving TUG (MD=-1.03, $P = 0.004$) and BBS (MD=2.93, $P<0.001$), and 20–45 min intervention, ≥3 times/wk, 5–8 wk cycles were significant in improving TUG (MD=-0.89, $P<0.001$; MD=-0.75, $P = 0.0003$; MD=-1.54, $P<0.0001$). VR intervention significantly improved TUG (MD=-2.27, $P<0.0001$) and BBS (MD=3.41, $P<0.0001$) in older adults in the hospital or nursing home compared with those residing in communities.

Conclusion: VR interventions can help the elderly with impaired balance to overcome traditional sports obstacles and improve physical function, balance and minimize falls. Balance training-based VR intervention is more effective in balance recovery and fall prevention compared with game program. An intervention plan comprising 20–45 min, 5–8 wk cycles, and ≥3 times/wk frequency has significantly higher effects for high-risk elderly populations living in hospitals or nursing homes.

1. Introduction

Balance is essential for older adults to perform daily activities. However, age-related decline in body function and balance increases the risk of falls and serious injury among the elderly. Some studies report that deterioration in joint proprioception and nerve connectivity inhibits accurate and timely postural adjustments in elderly patients, predisposing them to falls and even causes fractures and head injuries (Pereira, 2020). These injuries increase the hospitalization and mortality rate of the elderly as well as increases the medical burden of the family.

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https://doi.org/10.1016/j.archger.2023.104924
Received 5 October 2022; Received in revised form 5 December 2022; Accepted 4 January 2023
Available online 5 January 2023

Therefore, it is imperative to explore effective, feasible, and economical intervention measures to improve the physical activity and balance ability of the elderly to prevent falls and reduce mortality.

Previous studies report that regular physical exercise can delay the decline of physical function, improve balance ability, muscle strength, and other functional challenges associated with aging, which are evaluated with indicators such as promoting shorter timed up-and-go test (TUG), increased berg balance scale (BBS) scores, and improved falls efficacy scale (FES-I) (Liu et al., 2022; Meekes & Stanmore, 2017). Adhering to traditional physical activities or exercise can be challenging for elderly people living alone (Pacheco, T, 2020, Yen & Chiu, 2021, PED et al., 2021). Recent studies indicate that advances in science and technology can provide practical alternatives to traditional exercise programs. For example, Virtual reality (VR) -based game intervention has been widely used in physical exercise and health promotion for the elderly as a potential rehabilitation technique (Hall et al., 2016). VR-based interventions are becoming increasingly popular around the world with the development of digital health. VR technology can provide a real environment for simulating exercise and experience, allowing users to interact with multiple senses and receive real-time feedback (Mohammadi et al., 2019), and it is affordable and entertaining. Some commercial virtual game systems, such as the Nintendo Wii sports system, Microsoft Xbox (Kinect), and Dance Revolution System (Tong, 2016; Almajid et al., 2021; Yang et al., 2020), can detect changes in player activity and movement in the real world through wireless interface sensors, so that players can receive real-time feedback, action reflection, and reward in the 3D world to increase motivation and willingness to keep exercising. However, there is inconsistency and variability in selection criteria in previous studies, so there is still no consensus on the effectiveness of VR gaming interventions on physical function, balance, and falls in the elderly population. A study reported that the VR interventions group trained using virtual games did not exhibit better performance on TUG compared with the control group that received conventional treatment alone (Lee et al., 2014). In another study, Walker et al. (Walker et al., 2000) added visual feedback to the balance training intervention of patients with acute stroke, and the findings did not show significant difference between the two groups. The choice of virtual game intervention protocol, intervention frequency, and living environment (community, inpatient, and nursing facility) of the different patients in these studies may have contributed to the variability of the findings (Palacios-Navarro & Hogan, 2021). Significant improvement can be achieved by integrating and summarizing virtual game protocols to personalize each patient's intervention plan.

Therefore, the aim of this study was to evaluate the intervention effects of VR intervention among older adults with balance impairment and evaluate the effective intervention options for their balance promotion and physical function improvement. The findings of this study will provide evidence on the effectiveness of VR interventions for elderly patients to improve their balance and prevent falls and help elderly patients with impaired balance to optimize the selection of virtual gaming solutions.

2. Methods

2.1. Search strategy

Articles were retrieved from PubMed, Web of Science, Elsevier, Cochrane databases, and CNKI using the keywords "virtual reality game", "virtual reality intervention", "commercial games", "video games", "exergames", "old adults", "elderly", "aging", "physical function",

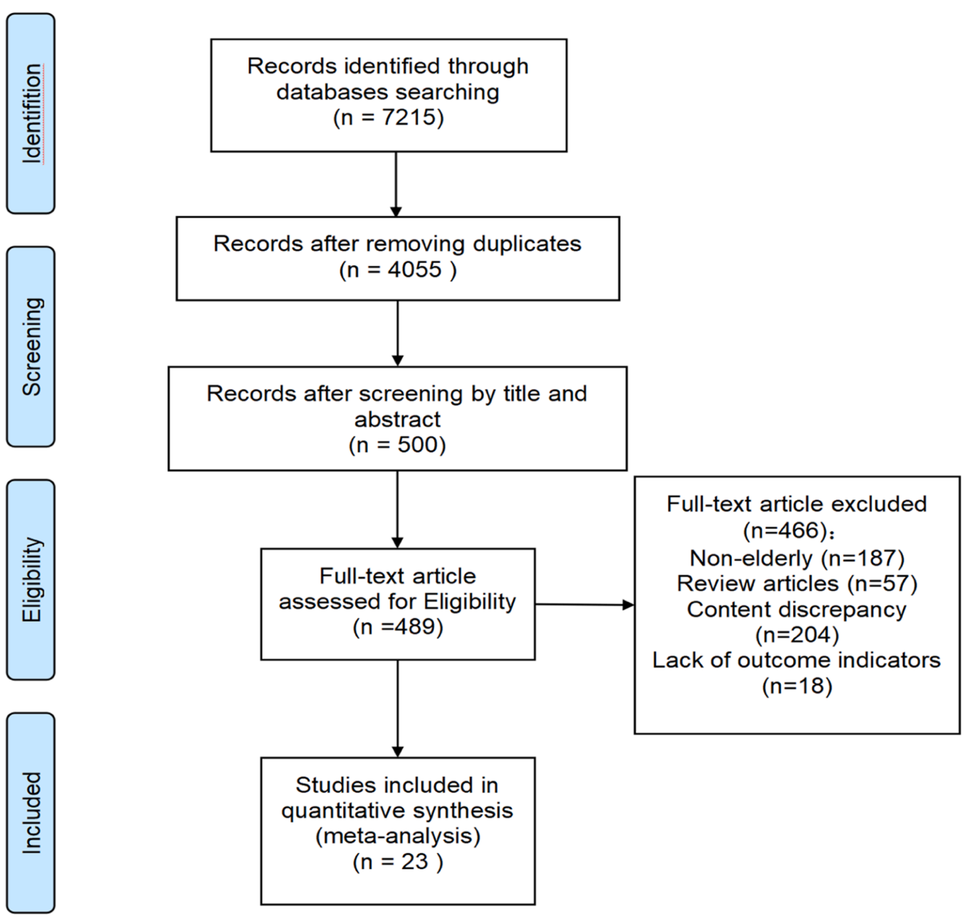

Fig. 1. Flow diagram showing inclusion and exclusion of studies.

Table 1
Characteristics of included studies.

Studies	Region	Participants Living	Age (Mean ±SD)	Sample Size (Male/ Female)	Intervention	Control	Duration, Frequency and Cycle	Outcomes
Agmon et al., 2011	Washington	Communities	84±5	7 (3/4)	Exergames (basic step, soccer heading, ski slalom, and table tilt)	No intervention	30 min, 3 times/wk, 12wk	BBS; 4-Meter Timed Walk test; GS
Daniel, 2012	USA	Senior centers and residential living centers	IG= 80 ±3.4/ 78.13 ±5.5; CG=72.6 ± 4.6	23 IG=8 (3/ 5)/ 8(3/5); CG=7 (3/ 4)	Wii-ft: games include bowling, tennis, and boxing	No intervention/ Seated exercise	45 min, 3 times/wk, 15 wk	SFT; ABC
Fakhro et al., 2019	Tyre and Saida districts	–	72.2 ± 5.2	60 IG=30; CG=30	Wii Balance Board	No intervention	40 min, 3times/wk, 8 wk	TUG; COP
Gomes et al., 2018	Brazilian	Hospital	IG=83.0 ± 5.9; CG=85.0 ± 6.2	30 IG=15; CG=15	Interactive video games (IVGs)	Received a booklet outlining the benefits and risks of physical activity	50 min, 2 times/wk, 7 wk	FGA; FES-I
Gschwind et al., 2015	Sydney, Australia	Community dwelling	IG=80.1 ± 6.3/ 82.5 ± 7.0; CG= 80.2 ± 6.5	124 IG=39/ 24; CG=61	Unsupervised exercise programs (KIN/ SMT systems)	No exergames	90 min, 1 time/wk, 16 wk	KES; Standing balance; FTSS; TUG
Jung et al., 2015	–	Senior citizen centers	IG=74.3 ± 2.1; CG=73.6 ± 2.4	16 IG=8; CG=8	Wii-Fit program (Wakeboard, Frisbee dog, jet ski, and canoe games)	No exercise	30 min, 2 times/wk, 8 wk	BBS; FRT; TUG; GS
Kwok et al., 2016	Singapore	Community dwelling	IG= 70.5 ± 6.7; CG=69.8 ± 7.5	8 IG=40 (4/ 36); CG=40 (8/ 32)	Wii Active exercise program	Traditional gym exercise program	60 min, 1 times/wk, 24 wk	Fall incidence; MFES; KES; TUG; GS; 6MWD; NCW
Liao et al., 2019	Taipei	Not mentioned	IG=79.6 ± 8.5; CG=84.1 ± 5.5	52 IG=27 (8/ 19); CG=25 (8/ 17)	Kinect_x005f_x0002_based exergaming group	Combined exercise	60 min, 3 times/wk, 12 wk	FRT; OLS
Meng et al., 2020	Sichuan	Community dwelling	IG= 69.3 ± 3.8; CG= 70.4 ± 3.1	40 IG=20 (8/ 12); CG=20 (9/ 11)	Virtual Situation Interactive Training (X5501-SK)	Healthy Education	30 –35 min, 5 times/wk, 12 wk	TUG; MFES
Merriman et al., 2015	Dublin and Belfast	Community dwelling	IG=74.1 ± 6.7; CG=73.4 ± 7.0	76 IG=38 (15/ 23); CG=38 (1/ 37)	Balance training intervention in virtual environments	Passive control condition	30 min, 2 times/wk, 5 wk	BBS; FES-I;SB; DB; SIFI
Mirelman et al., 2016	Israel	Community	IG= 74.2 ± 6.9; CG=73.3 ± 6.4	282 IG=146 (98/48); CG=136 (84/52)	Treadmill training plus VR (including obstacles, multiple pathways and distractors)	Treadmill training	45 min, 3 times/wk, 6 wk	GS; 2MWD; SPPB; PAS; SF-36
Molhemi et al., 2020	–	–	IG=36.8 ± 8.4; CG=41.6 ± 8.4	39 IG=19 (7/ 12); CG=20 (8/ 12)	VR (Xbox360 with 140 Microsoft's Kinect)	Conventional balance training	30 min, 3 times/wk, 6 wk	TUG; 10 MWT; BBS; FES-I; ABC
Montero-Alía et al., 2019	Spain	Community dwelling	IG=75.1 ± 1.3; CG=75.4 ± 1.7	977 IG=508 (192/316); CG=469 (208/261)	Balance training (NintendoTM Wii console and the WBB, balance bubble)	Usual care	30 min, 2 times/wk, 48wk	Tinetti balance test; Unipedal stance; Wii balance tests; FES-I; FGA
Padala et al., 2017	Central Arkansas	Community dwelling	IG=67.5 ± 8.1; CG=69.0 ± 3.8	30 IG=15 (13/ 2); CG=15 (13/2)	Wii-Fit program (Yoga, Strength training)	Computer-based cognitive program	45 min, 3 times/wk, 8 wk	BBS; ABC; SF-36
Park and Yim, 2016	Seoul	Community-dwelling	IG=73.0 ± 3.0; CG=74.1 ± 2.9	72 IG=36 (3/ 33); CG=36 (1/ 35)	3-D virtual reality Kayak program (paddling)	No intervention	20 min, 2 times/wk, 6 wk	Balance; Arm curl; Muscle strength
Phu et al., 2019	Albans	Community dwelling	IG=79; CG=79	195 IG=63 (19/	Virtual reality training (rehabilitation exercise) using	No intervention/ Exercise using a		FTSS; TUG; GS; FES-I; HGS

(continued on next page)

Table 1 (*continued*)

Studies	Region	Participants Living	Age (Mean ±SD)	Sample Size (Male/ Female)	Intervention	Control	Duration, Frequency and Cycle	Outcomes
				44)/ 82 (31/51); CG=50 (15/35)	the Balance Rehabilitation Unit (BRU)	modified Otago Exercise Program	45 min, 2 times/wk, 6 wk	
Santos et al., 2019	Brazil	Community dwelling	IG= 69.1 ± 5; CG=69.7 ± 5.6	20 IG= 9; CG=11	X-box 360 game Your Shape Fitness Evolve	Resistance training	40 min, 3 times/wk, 12 wk	FTSS; TUG; 10 MWT
Sun et al., 2017	Nanjing	Nursing homes	IG= 78.0 ± 4.3; CG= 77.7 ± 3.9	60 IG=30 (12/ 18); CG=30 (11/19)	VR training and rehabilitation system	No intervention	50 min, 3 times/wk, 12wk	TUG; 30 s CST; MFES; DB
Stanmore et al., 2019	Manchester, Glasgow	Assisted living	IG=77.9 ± 8.9; CG=77.8 ± 10.2	106 IG=56 (11/ 45); CG=50 (12/ 38)	Balance Exergame program	Standard care	30 min, 3 times/wk, 12 wk	TUG; BBS; FES-I; FRAT; PAS
Szturm et al., 2011	–	Community dwelling and ambulatory	IG=80.5 ± 6.0; CG=81.0 ± 7.0	30 IG=15 (5/ 10); CG=15 (6/ 9)	Dynamic balance exercises coupled with video game play	Typical rehabilitation program consisting of strengthening and balance exercises	45 min, 2 times/wk, 8 wk	BBS; ABC; TUG; GS
Tsang & Fu, 2016	Hong Kong	Nursing homes	IG=82.3 ± 3.8; CG=82.0 ± 4.3	79 IG=39 (16/ 23); CG=40 (15/25)	Wii Fit balance training (Soccer Heading, Table Tilt, Balance Bubble)	Conventional balance training	60 min, 3 times/wk, 6 wk	BBS; TUG
Yeşilyaprak et al., 2016	Turkey	Nursing homes	IG= 70.1 ± 4; CG=73.1 ± 4.5	18 IG=7 (3/ 4); CG=11 (9/2)	Balance training with the BTS Nirvana VR Interactive System	Conventional exercise	35–45 min, 3 times/wk, 6wk	BBS; TUG; OLS; FES-I; Tandem stance
Zahedian-Nasab et al., 2021	Iran	Nursing homes	IG= 69.7 ± 7.7; CG=72.0 ± 7.8	60 IG=30 (22/ 8); CG=30 (22/8)	Exercise by Xbox Kinect	Routine programs	35–45 min, 2 times/wk, 6 wk	BBS; TUG; FES-I

Note: IG= Intervention group; CG= Control group; GS=Gait speed; KES=Knee extension strength; HGS=Hand grip strength; ABC=Activities-specific confidence scale; MFES=Modified falls efficacy scale; COP:Center of pressure; FTSS=Five times sit-to-stand test; FRT=Functional reach test; 6MWD=6-minute walk test; 10 MWT=10-meter walk tests; SPPB=Short physical performance battery and score sheet; FGA=Gait assessment; NCW=Narrow corridor walk test; PAS=Physical activity scale; SFT=Senior fitness test; SF-36=Rand short form 36; 30sCST=30-second chair stand test; FRAT=Fall risk score; OLS=One-leg stance test; SB=Static balance function; DB=Dynamic balance function.

"balance", and "fall" Articles published from inception of the database to March 20, 2022 were retrieved. The search scope was limited to English and Chinese articles.

2.2. Inclusion and exclusion criteria

Inclusion criteria: 1) Elderly participants with impaired balance including, subjects with Berg Balance Scale (BBS) scores lower than 52 (indicating impaired balance) (Berg et al., 1992) or those with a history or risk of falls or weak muscle strength and balance. 2) Randomized controlled trials (RCT) and case comparison studies were included. Studies with VR game intervention as the VR intervention group and non-intervention group or regular training or physical activity as the control group were included in this study. 3) Studies with outcome indicators including physical function, balance, and falls were selected. Studies reporting one or more of the relevant indicators were eligible for inclusion in this study.

Exclusion criteria: 1) Studies that comprised subjects that were not elderly or enrolled elderly patients with cancer, degenerative neurological diseases, major depression, etc. were excluded. 2) Review and systematic analysis studies or experimental studies that did not include VR interventions were excluded. 3) Experimental studies with other outcome indicators that are not among the ones mentioned above were excluded.

2.3. Research selection and data extraction

Two authors initially selected research papers according to the titles and abstracts. The articles that met the inclusion criteria were then selected for full-text reading and targeted analysis. All steps were carried out independently by two investigators, and a third investigator (Lu) was involved when the first two authors disagreed or could not reach a decision to help in reaching a consensus. The extracted data included: author, year of publication, region, basic information of subjects (sample size, age, gender, and residential location), VR game intervention (type, duration, frequency, and cycle), outcome indicators (physical function, balance, and self-rated fall score).

2.4. Literature quality assessment

The "Risk of Bias tool" in the Cochrane Manual was utilized to evaluate the risk of bias and the quality of the articles. Quality assessment was conducted from six aspects: random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, and selective reporting.

2.5. Statistical analysis

Descriptive statistics were used to describe the extracted data. The

Table 2
Risk of bias assessment of included studies.

References	Random sequence generation	Allocation conceal ment	Blinding of participants and personnel	Blinding of outcome assess ment	Incomplete outcome data	Selective reporting	Other sources of bias
Agmon et al., 2011	low risk	low risk	high risk	unclear risk	low risk	unclear risk	unclear risk
Daniel, 2012	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Fakhro et al., 2019	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Gomes et al., 2018	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk
Gschwind et al., 2015	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Jung et al., 2015	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk
Kwok et al., 2016	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk
Liao et al., 2019	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Meng et al., 2020	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk
Merriman et al., 2015	low risk	low risk	low risk	unclear risk	unclear risk	unclear risk	unclear risk
Mirelman et al., 2016	unclear risk	low risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Molhemi et al., 2020	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk
Montero-Alía et al., 2019	low risk	low risk	low risk	low risk	low risk	unclear risk	low risk
Padala et al., 2017	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Park et al., 2016	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk
Phu et al., 2019	low risk	low risk	low risk	unclear risk	low risk	unclear risk	unclear risk
Santos et al., 2019	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Stanmore et al., 2019	unclear risk	low risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Sun et al., 2017	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Szturm et al., 2011	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Tsang et al., 2016	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk
Yeşilyaprak et al., 2016	unclear risk	unclear risk	unclear risk	unclear risk	low risk	unclear risk	unclear risk
Zahedian-Nasab et al., 2021	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk	unclear risk

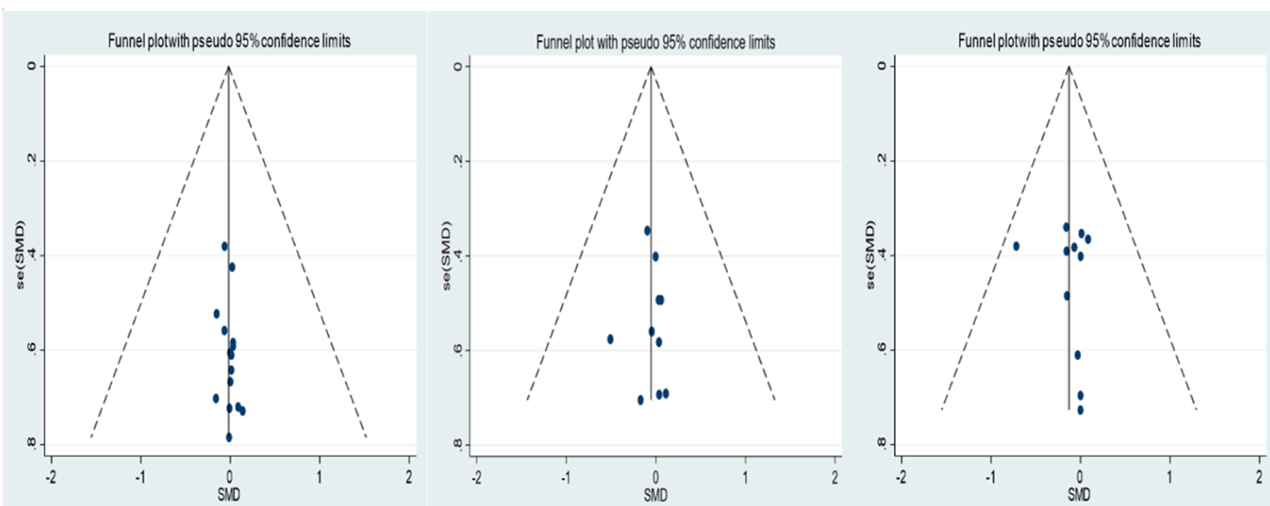

Fig. 2. Funnel plot for TUG, BBS and FES-I.

standardized mean difference (SMD) or mean difference (MD) was calculated (95% CI) because the retrieved data was continuous. The heterogeneity among the results was determined using I^2. A higher I^2 indicated significant heterogeneity between studies. The fixed-effect model was adopted if the heterogeneity among the results was not significant ($I^2 < 50\%$); whereas the random effect model ($I^2 > 50\%$) was adopted when the heterogeneity was significant. Subgroup analysis and sensitivity analysis were performed to test the reliability of the results. A funnel plot was used to evaluate publication bias of the studies. The statistical significance was considered at $\alpha = 0.05$.

3. Results

3.1. Literature search results and basic characteristics of included studies

A total of 7215 articles were retrieved. The documents were initially screened by reading the titles and abstracts, and the selected articles were further manually reviewed. Finally, 23 articles were included in this study (Fig. 1).

The type of VR game equipment in the included studies mainly included Wii sports and Xbox systems, and VR interventions were mainly based on balance training or on game programs. Participants included in 14 studies lived in the community or from a senior center, subjects in 6 studies lived in a hospital or nursing facility, and 3 studies did not report the residential setting of the participants. Notably, 10 of the included articles mentioned the presence of a supervisor during the VR interventions for older adults, whereas the remaining 13 articles did not report whether a supervisor was involved (Table 1).

3.2. Literature quality and publication bias risk

"High risk", "low risk" and "unclear" were used to evaluate each index in the included literature, and the results are presented in Table 2.

Stata software was used to construct a funnel plot of the TUG, BBS, and FES-I results (Fig. 2). Egger's test was conducted to evaluate publication bias risk. The funnel plot showed satisfactory symmetry

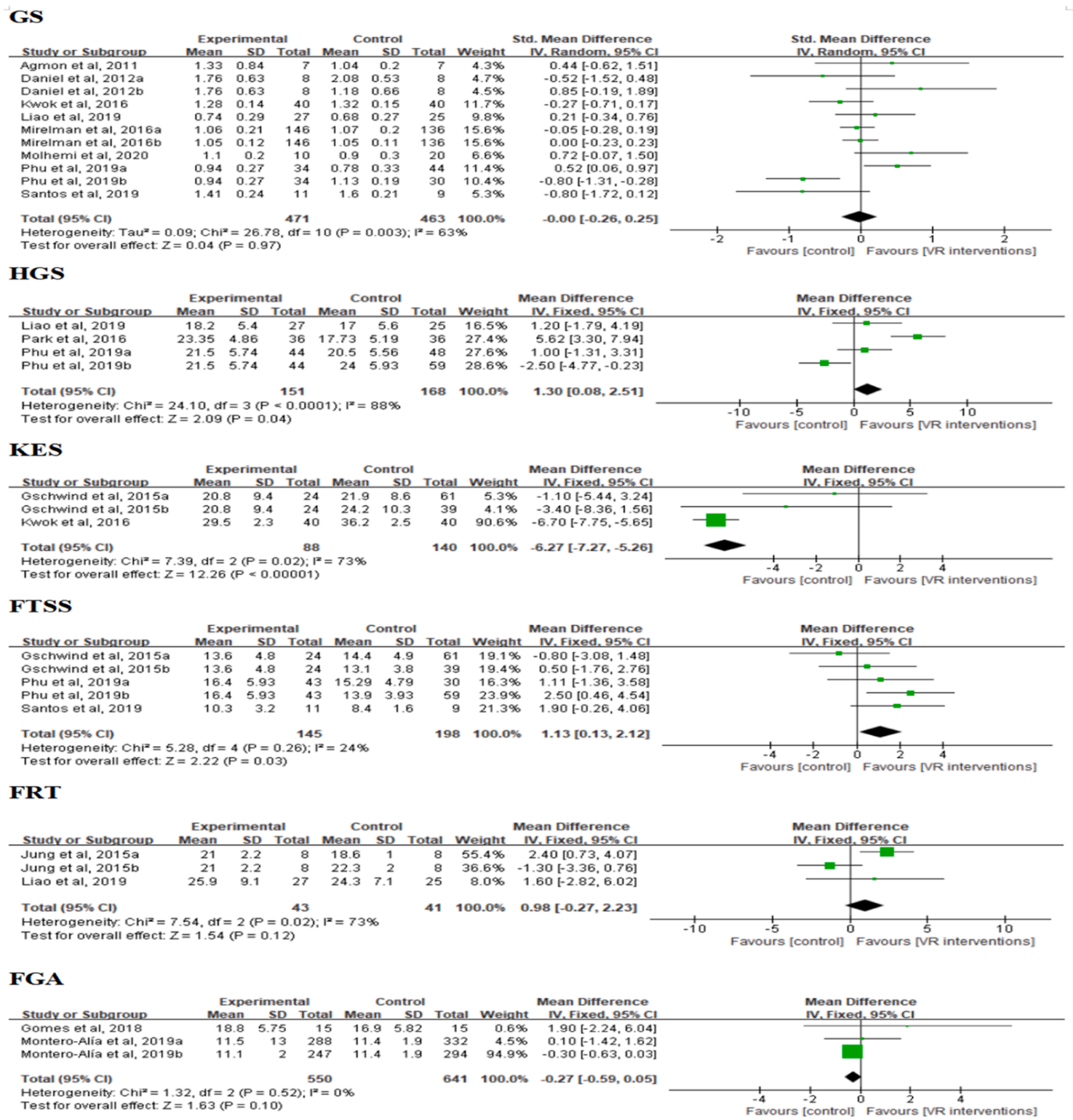

Fig. 3. Forest plot of effect of VR interventions on physical function.

between the left and right sides. Egger's test showed no significant publication bias in TUG ($Z = 0.13$, $P<0.001$, $t = 0.220$, $P = 0.900$), BBS ($Z = 0.330$, $P<0.001$, $t = 0.860$, $P = 0.741$) and FES-I ($Z = 1.010$, $P<0.001$, $t = 3.130$, $P = 0.313$).

3.3. Effects of VR interventions on physical function, balance and fall risk

The physical function indicators reported in the studies included GS, HGS, KES, FTSS, FRT and FGA. A total of 12 articles compared GS between the VR interventions group and the control group (non-intervention or conventional exercise group), and 4 studies compared HGS between the two groups, 5 articles compared FTSS between the two groups, and 3 studies separately compared KES, FRT and FGA indicators between the two groups (Fig. 3).

The results showed significant differences in HGS, KES and FTSS between the VR interventions group and the control group. The HGS (MD: 1.30; 95% CI: 0.08, 2.51; $P = 0.040$), KES (MD: −6.27; 95% CI: 7.27, 5.26; $P<0.001$) and FTSS (MD:1.13; 95% CI: 0.13, 2.12; $P = 0.030$) of VR game intervention group were significantly higher than the control group. The meta-analysis results showed no significant difference in GS between the VR interventions group and control groups ($P = > 0.05$). Moreover, there was no significant differences in FRT ($P = > 0.05$) and FGA ($P = > 0.05$) indices between the two groups, possibly due to the low number of included studies.

Indicators used to evaluate balance and fall risk included TUG, BBS, ABC, and FES-I. A total of 21 studies compared the TUG scores between the VR interventions group and the control group (conventional exercise or non-intervention group), 11 studies compared BBS indicators between the two groups, 12 studies compared FES-I indicators between the two groups, and 6 studies compared ABC indicators between the two

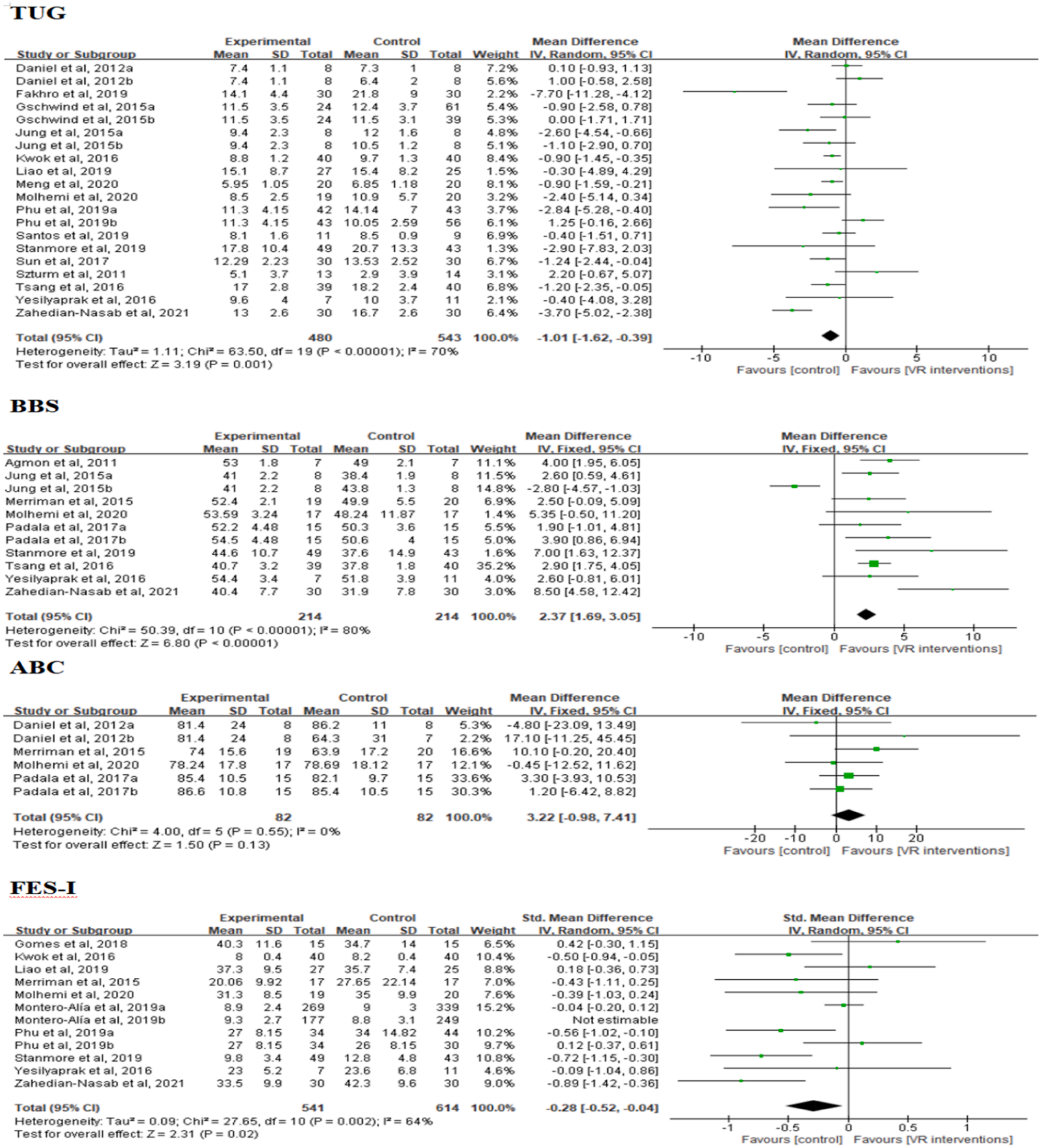

Fig. 4. Forest plot of the effect of VR interventions on balance and fall risk.

groups. The results showed significant differences in TUG, BBS and FES-I between the VR interventions group and control groups. The TUG score (MD:−1.01; 95% CI:−1.62,−0.39; P = 0.001) and FES-I score (SMD:−0.28; 95% CI:−0.52,−0.04; P = 0.020) in the VR interventions group were significantly lower than the control group. The BBS score of subjects in the VR interventions group was significantly higher than the control group (MD:2.37; 95% CI:1.69, 3.05; P<0.001). The results on these indicators did not show significant differences in ABC scores between the two groups. Notably, a significant heterogeneity was observed among the studies (I^2=65%, P<0.001; I^2=80%, P<0.001; I^2=64%, P = 0.002), and the heterogeneity among studies did not decrease significantly after excluding each study at a time (Fig. 4). Subgroup analysis was then conducted to achieve significant reduction in the heterogeneity among studies.

3.4. Subgroup analysis

3.4.1. Subgroup analysis based on different control groups

In this study, subgroup analyses were performed according to the control group categories (conventional exercise and non-intervention). Conventional exercise was defined as use of an intervention that comprised traditional exercise methods in the control group. Subgroup results showed that the VR intervention was more effective on TUG and BBS than the conventional exercise group(MD=−0.54, P = 0.004;

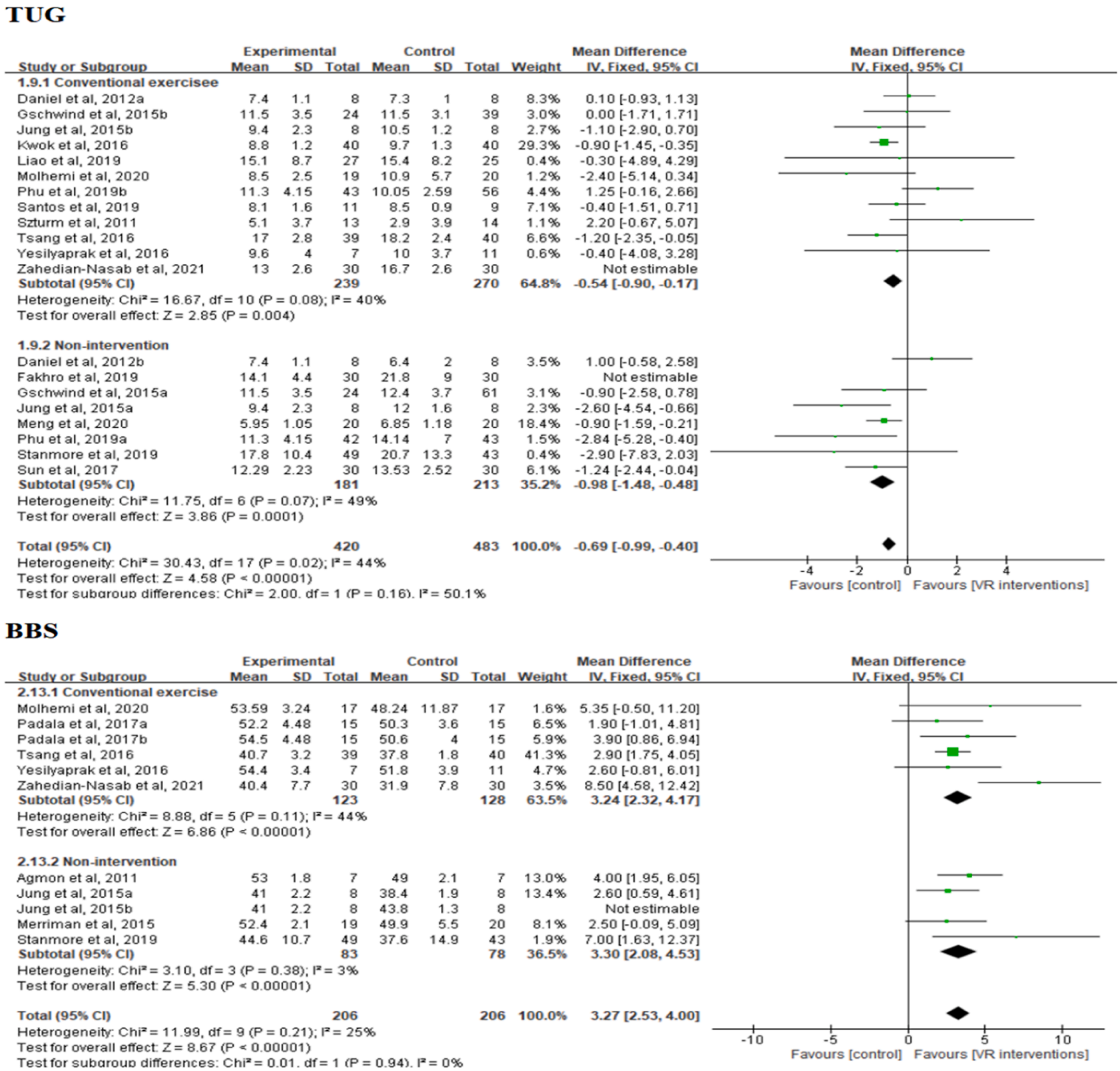

Fig. 5. Subgroup analysis based on different control groups.

MD=3.24, P<0.001) and the non-intervention group(MD=−0.98, P = 0.001; MD=3.30, P < 0.001) (Fig. 5).

3.4.2. Subgroup analysis based on different types of VR interventions

Subgroup analysis based on different VR interventions (balance training, game program) was conducted for the TUG and BBS indicators. The results showed a significant difference in the effect of the two VR interventions on TUG and BBS scores. Balance training had a significant effect on improving TUG (MD=−1.03, P = 0.004) and BBS (MD=2.93, P<0.001) than the game program (MD=2.85, P = 0.004) (Fig. 6).

3.4.3. Subgroup analysis based on different intervention duration, frequency and cycle

In this study, subgroup analyses were conducted based on different intervention durations (20–45 min, >45 min), frequencies (<3 times/wk, ≥3 times/wk), and cycles (5–8 wk, >8 wk) (Fig. 7). The results of subgroup analysis for the different intervention duration showed that the 20–45 min intervention duration had significantly improved TUG score (MD=−0.89, P<0.0001) and FES-I score (MD=−0.28, P = 0.030), whereas > 45 min intervention duration had no significant effect on FES-I score (P => 0.05). The effect of the 20–45 min intervention duration on TUG score was higher(MD=−0.89, P <0.001) compared with the effect of >45 min intervention duration (MD=−0.88, P =0.0002).

Subgroup analysis based on the intervention frequency showed that ≥3 times/wk intervention had a significant intervention effect on improving TUG (MD=−0.75, P = 0.0003), BBS (MD=3.22, P<0.0001), and FES-I (SMD=−0.56, P = 0.0009) indicators, whereas <3 times/wk intervention frequency only had a significant intervention effect on improving the TUG indicator (MD=−1.02, P<0.0001) (Fig. 8).

Only few studies included in the analysis reported the >12 wk intervention frequency, so subgroup analysis was only performed for the 5–8 wk and >8 wk intervention durations in this study. The results showed that intervention cycles of the 5–8 wk intervention had a positive effect on TUG (MD=−1.54, P<0.0001), BBS (MD=3.11, P<0.0001), and FES-I (SMD=−0.40, P = 0.010) indicators. The >8 wk intervention had a significantly higher effect on BBS (MD=3.69, P<0.0001) than the 5–8 wk intervention (MD=3.11, P<0.0001) (Fig. 9).

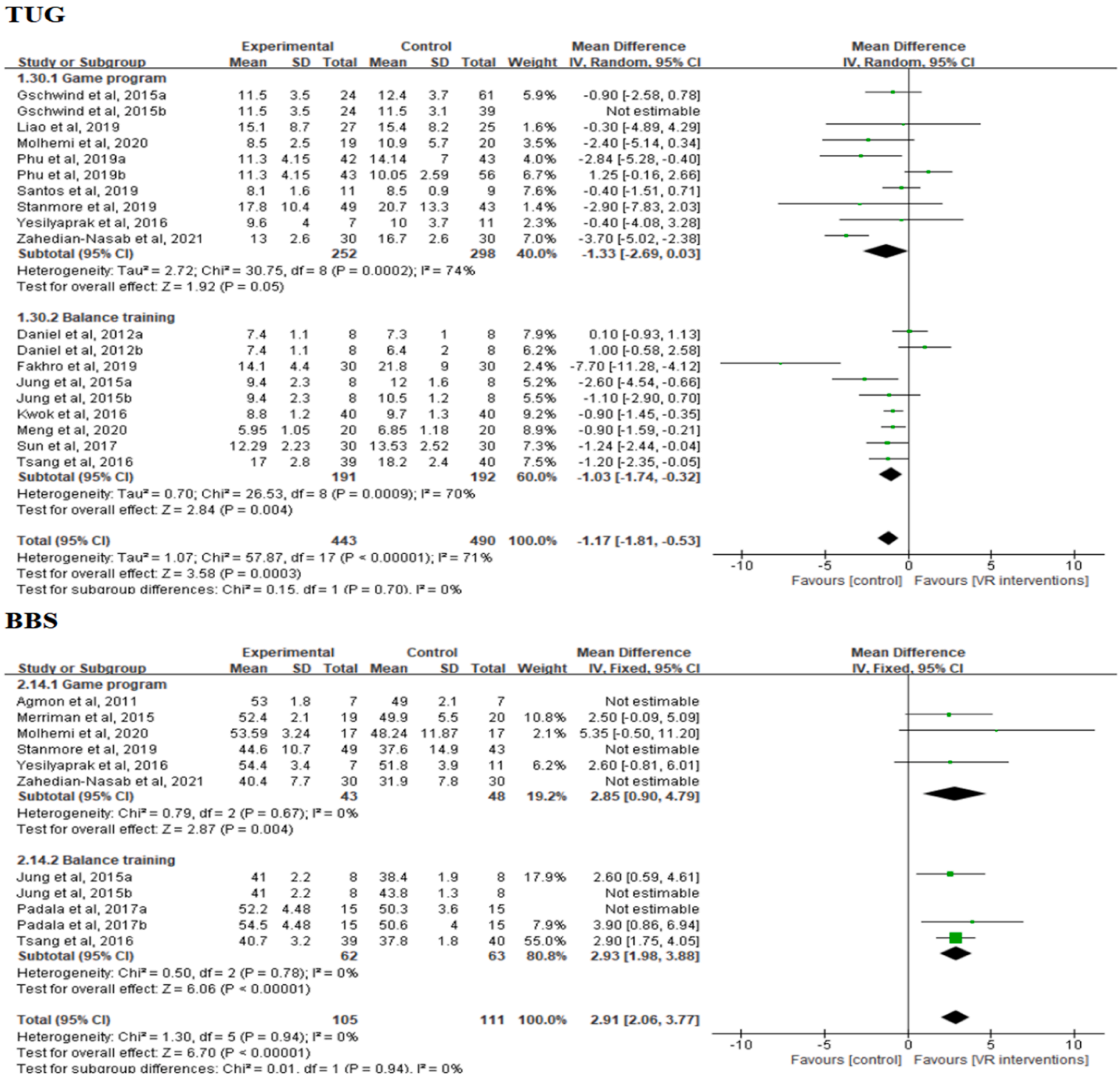

Fig. 6. Subgroup analysis based on different VR interventions types.

3.4.4. *Subgroup analysis based on the residential locations of subjects*

Subgroup analysis was conducted based on the different residential locations of the elderly subjects. The results showed that the VR intervention had a significantly different effect on the TUG and BBS of elderly people in different residential settings (hospital or nursing home, community-dwelling). The VR intervention significantly improved TUG (MD=−2.27, $P<0.0001$) and BBS (MD=3.41, $P<0.0001$) for elderly subjects in the hospital or nursing home than compared with subjects living in the community, whereas there was no significant effect of VR intervention on the FES-I indicators of all subjects ($P>0.05$) (Fig. 10).

4. Discussion

4.1. *Major findings*

The meta-analysis results showed that the VR intervention improved physical function indicators such as HGS, KES, and FTSS in balance-impaired elderly subjects. Positive effects for balance ability indicators, were only observed in TUG and BBS indicators. Positive effects of these indicators indicated improvement in walking function or reduction in possible risk of falling during walking. This implies that elderly patients with impaired balance exhibited positive results in terms of physical function, balance, and fall risk. Previous findings indicate that VR intervention can be used rehabilitative purposes to improve balance and gait ability in elderly patients (Erhardsson et al., 2020; Komagata et al., 2021). VR interventions reduce the risk of falls in older adults (Choi et al., 2017), as well as alleviate depressive symptoms (Li et al., 2021) and improve quality of life of these subjects (Li et al., 2021; Cugusi et al., 2021). Previous findings indicate that VR interventions are effective in improving the walking ability of older adults with neurological disorders (Janhunen et al., 2021). The ecological, safe and motivational treatments offered by virtual games are effective in promoting rehabilitation in older adults. This effect can be attributed to the advantages of VR games such as its participatory and interactive nature. Virtual gaming environments focus on guiding the player through a realistic experience, which improves the concentration of elderly patients as well as promotes their cognitive and executive function recovery. In addition, the immersive gaming experience helps

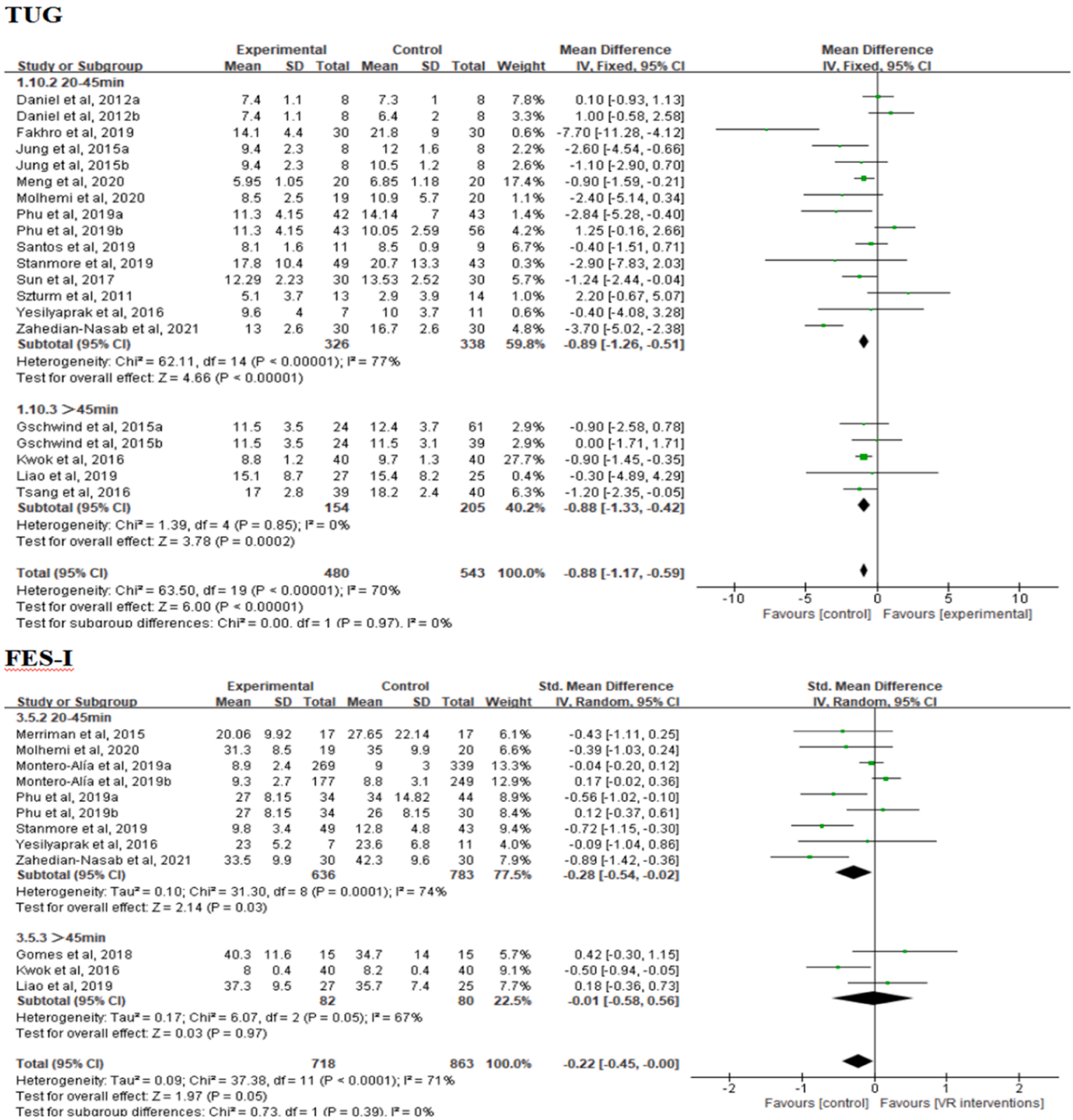

Fig. 7. Subgroup analysis based on intervention duration.

older patients recover from cardiovascular and respiratory, metabolic and immune system disorders (Morrison et al., 2018; Mpt M, 2020).

The present results show that studies with high heterogeneity exhibited significant effects of VR interventions, thus ubgroup analyses were further performed based on the different control groups, different types of VR games, different intervention doses, and different residential locations of the subjects, to examine the sources of heterogeneity for each outcome with significant differences. The results of the subgroup analysis showed that VR interventions were significantly more effective in improving TUG and BBS scores in older adults than the conventional exercise and non-intervention group. This is consistent with previous findings that virtual games have more significant effects than conventional physical exercise in strengthening lower limb muscles, improving flexibility, and counteracting age-related decrease in neuromotor and somatosensory functions (Mohammadi et al., 2019). The benefits of VR interventions are also reported in several other studies (Hashemi et al., 2022; Leonardi et al., 2021). Some studies report that lower adherence to traditional exercise and physical activity in older adults can be attributed to lack of motivation, sports phobia, and other factors (Pacheco, 2020). Conventional exercise programs are considered monotonous and boring, thus loss of interest is a common reason for the elderly to terminate their regular exercise program. However, in VR games, player participation requires active mobilization of various body parts to maintain interaction, and the interactive simulation reflects the daily virtual environment and allows subjects to repeat physical exercises that are directly related to the movements required for daily life activities. VR games have close resemblance to reality thereby the elderly can actively interact with the game scenario during the game play. This continuously stimulates the participants physical sensory, cognitive, psychological, and motor functions (Maggio et al., 2019) and

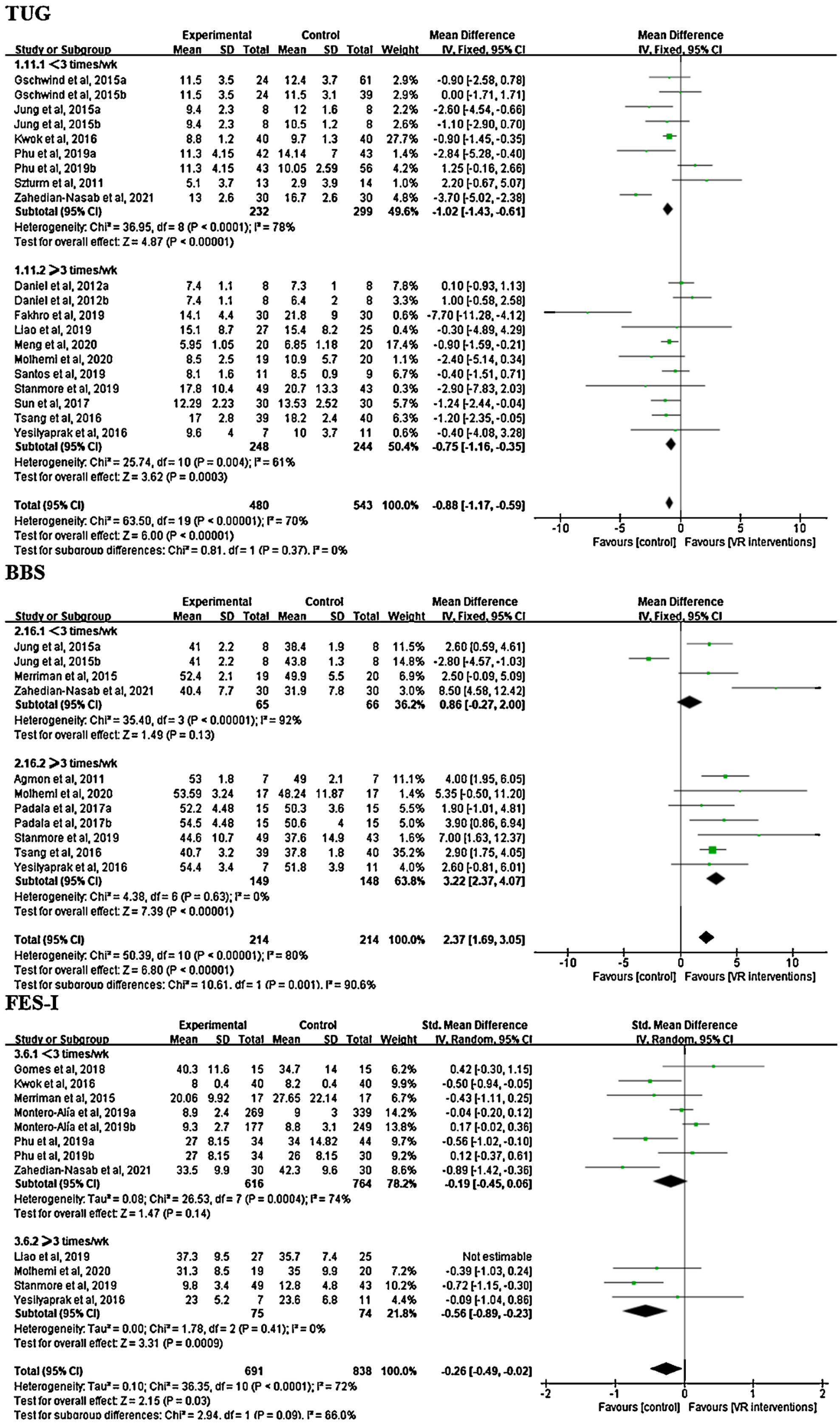

Fig. 8. Subgroup analysis based on intervention frequency.

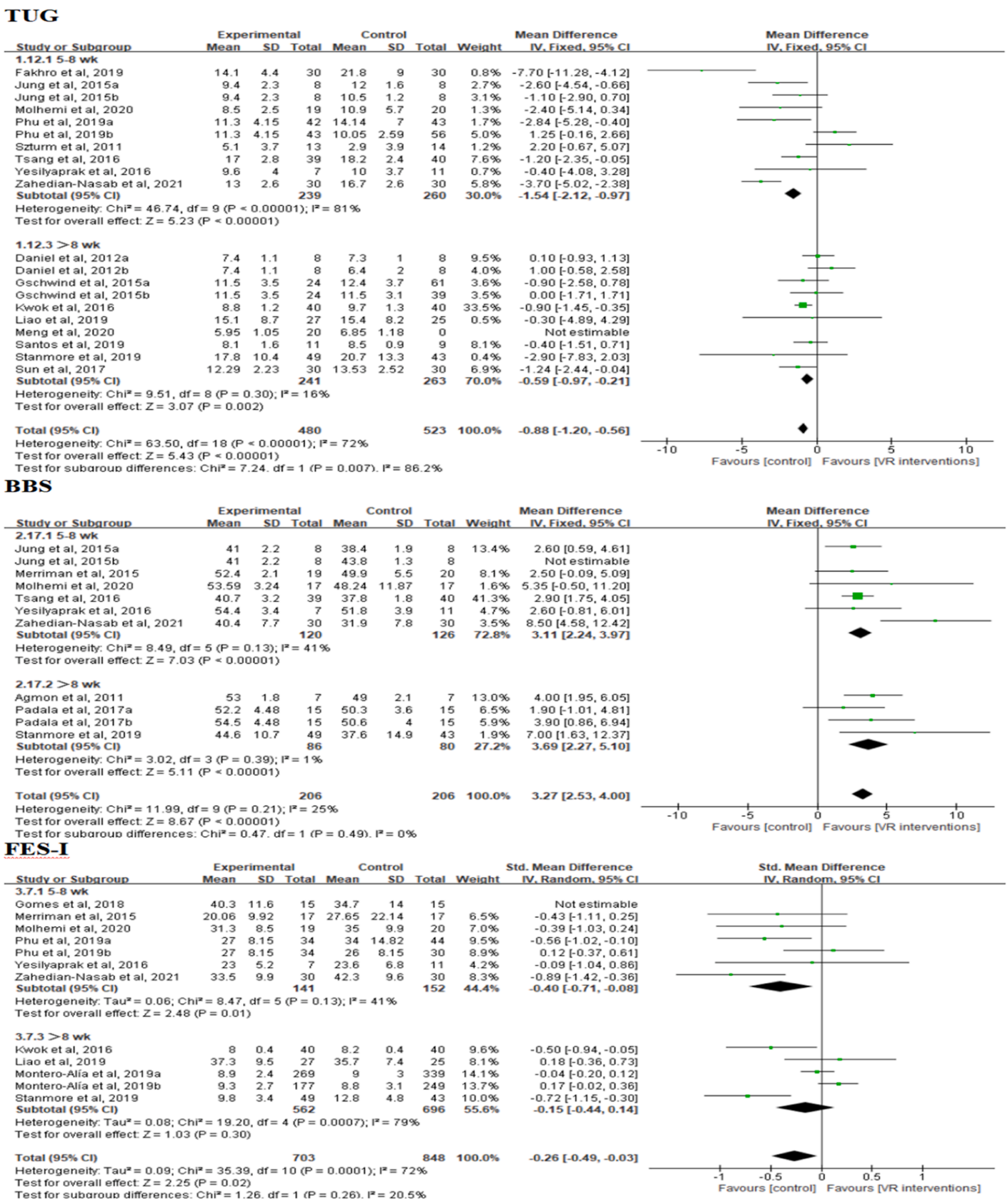

Fig. 9. Subgroup analysis based on intervention cycle.

increases their interest in practicing physical exercise in an immersive and attractive environment (Bettger, 2020; Wen et al., 2021). These exercise environments are also beneficial for improvement of older adults' compliance and can help them overcome some of the traditional disadvantages of exercise barriers (Wiemeyer & Kliem, 2012), such as lack of motivation and negative perceptions of exercise outcomes. VR intervention is a fun and engaging game with controlled participation and interaction by physical movement in a safe environment, which has several advantages over traditional exercise for the promotion of motor function in the elderly.

In this study, inconsistent positive effects were observed among the different VR intervention types. The findings showed that balance

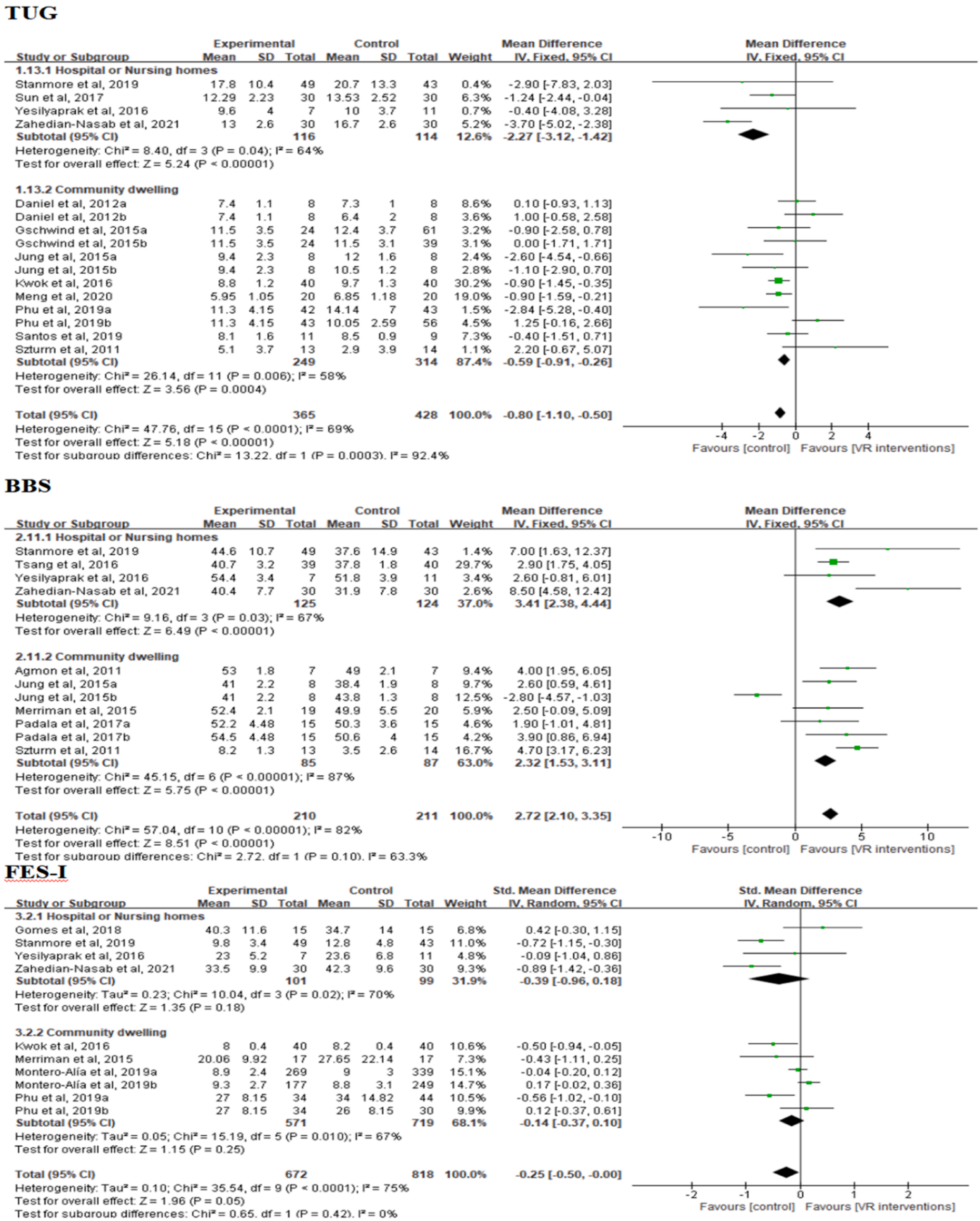

Fig. 10. Subgroup analysis based on residential location for the elderly.

training-based VR interventions were significantly more effective in improving TUG and BBS score than game program-based VR interventions. This implies that appropriate selection of these interventions is critical in improving balance control and reducing the risk of falls in older adults with balance disorders. The VR interventions can respond to changes in the frequency, direction, velocity, and acceleration of motor movements, encouraging players to complete task sets and achieve desired goals (Meekes & Stanmore, 2017), which ensures the safety and effectiveness of virtual game interventions. The results of subgroup analysis based on different intervention durations showed that the 20–45 min intervention duration had a significant effect on improving TUG than the >45 min intervention duration, and the 5–8 wk intervention cycle was more effective in improving TUG, whereas the >8 wk intervention cycle had a better intervention effect on BBS. The intervention effect of ≥3 times/wk intervention frequency had a markedly high improvement effect on TUG, BBS, and FES-I indexes compared with the other frequencies. This indicates that the plan with 20–45 min intervention duration, ≥3 times/wk intervention frequency and 5–8 wk intervention cycle exhibited the highest effect on improving some of the outcomes. Elderly patients with different individual differences should be grouped according to their age, health status, and disease severity to further determine the appropriate intervention program to achieve greater improvement. Notably, the analysis based on the dose of different exercise interventions was not further explored due to the low number of patients included in the articles. Therefore, more VR game intervention studies should be conducted to explore the effects of different intervention doses.

Furthermore, the results of the subgroup analysis based on different residential locations showed that the VR game intervention had a better effect on the improving TUG and BBS indicators in elderly subjects living in hospitals or nursing homes than in those living in the community-dwelling, whereas there was no significant effect of the intervention on FES-I indicators. This implies that the VR game intervention may exhibit more benefits in terms of enhanced balance and motor function in balance-impaired older adults, but the improvement effect on self-rated fall efficacy was not significant. The current results indicate the need for an enhanced VR gaming intervention to improve balance in elderly people with balance impairment to prevent their physical activity function from declining, especially in the high-risk elderly population residing in hospitals or nursing homes.

4.2. Advantages and limitations

In this study, the effects of various VR interventions on physical function, balance, and fall efficacy in older adults with balance impairment were systematically evaluated. The optimal exercise plans for promoting balance function in the elderly with balance impairment were evaluated. The results of this study can help in establishing new and more effective methods of VR game support or combination to explore more specific interventions, but the study had some limitations. First, some of the results in the included articles provide pre-intervention and post-intervention information in percentage terms, thus the effect sizes could not be compared with other studies. Although we contacted the authors of the articles, we may have missed some information. Second, more definitive conclusions cannot be drawn due to insufficient literature and major differences between intervention programs and measurement results. Therefore, studies should be conducted to explore more applications of VR intervention among the elderly, and combine more detailed, programmed, and scientific interventions to manage balance promotion and prevent falls in older adults with impaired balance.

5. Conclusion

The effect of VR game intervention on physical function and balance ability of the elderly with impaired balance was better than that of conventional exercise or non-intervention training. A balance training-based VR game can be recommended because of its better effect compared with other VR game type reported in the studies. The 20–45 min duration intervention plan, 5–8 wk, and ≥3 times/wk cycle were more effective in improving balance of older adults with impaired balance, especially in high-risk elderly populations living in hospitals or nursing homes.

Funding

This work was supported by the Innovation Project of Postgraduate Training in Jiangsu Province Bureau (no. KYCX21_2909) awarded to Yuanyuan Ren.

Author statement

All authors contributed to the conceptualisation of the paper. Yuanyuan Ren proposed the thesis theme and wrote the paper; Chenli Lin collected literature; Qin Zhou and Zhang Yingyuan revised the paper and verified the data; Guodong Wang and Aming Lu designed the thesis and improved the paper framework. All authors saw and approved the final version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships in this paper.

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